The block-rule Laurent cascade at $n = 9$:

empirical verification of the all-n locality engine

(satellite for the $\mathrm{Tr}(\phi^3)$ locality program)

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Abstract

At n external legs, after Step-1 (Layer-0) and Step-2 (bare-swap) exhausts have been applied, a residual set of non-local coefficients survives. Through $n = 8$ this residual was killed by a *depth-1 Laurent cascade* on each cyclic orbit independently: each orbit admitted a fingerprint equation whose cousins were Step-1-killable. At $n = 9$ we exhibit, for the first time, a coupled cluster of 90 orbits in which no individual orbit admits a single-orbit cascade recipe. The cluster's depth-1 fingerprint matrix $M^{\mathcal{C}}_{\mathrm{FP}}$ has 506 rows and rank exactly $90 = |\mathcal{C}|$, residual 0, so the entire cluster is killed *jointly* by the depth-1 system. Combined with single-orbit kills on the remaining 23 orbits, every Step-1+2 survivor at $n = 9$ is forced to zero. This generalises the $n = 6$ perfect-matching block (rank 3 on 4 orbits, residual 1, closed by an anchor at X_*^1) to a setting where the equation count overdetermines the cluster and no anchor is required. We refine Conjecture 10 of the master plan accordingly: the all-n engine is the *block-rule* on coupled clusters, not single-orbit recipes.

1 Background: the three layers of kill mechanism

We adopt the conventions of the root README. The cyclic hidden-zero zone $\mathcal{Z}_{r,r+2}$ has special chord $X_{r,r+2} =: X_*$, bare chord $X_{r+1,r-1} = -X_*$, and substitutions $X_{r+1,k} = X_{r,k} - X_*$ for k in cyclic range. The general ansatz is $B = \sum_M a_M / \prod_{c\in M} X_c$ over size-$(n-3)$ chord multisets M.

Definition 1 (Three layers)**.**

- **Step-1 (Layer-0)**: send $X_* \to \infty$ at each $\mathcal{Z}_{r,r+2}$. The leading-order coefficient identity forces $a_M = 0$ when $M \in \mathcal{K}_{r,r+2}$, i.e. when M contains neither special nor bare and no (companion, substitute) pair. Holds for M killable at *any* cyclic zone.
- **Step-2 (equate)**: keep X_* in the surviving variables and exploit $X_{r+1,r-1} = -X_*$. Pairs of multisets differing by a bare/special swap are glued, yielding identities $a_{M\cup\{X_{r+1,r-1}\}} = -a_{M\cup\{X_*\}}$. Cyclic shifts produce equalities and trivial cancellations.
- **Step-3 (Laurent cascade)**: at zone $\mathcal{Z}_{r,r+2}$, expand

$$\frac{1}{X_{r+1,k}} = \frac{1}{X_{r,k} - X_*} = -\frac{1}{X_*} - \frac{X_{r,k}}{X_*^2} - \frac{X_{r,k}^2}{X_*^3} - \cdots$$

  to subleading order. The coefficient of a free-variable monomial (the *fingerprint*) at order $X_*^{-(\ell_r(M)+1)}$ is a linear identity among coefficients a_M of multisets contributing the same fingerprint.

Definition 2 (Cyclic orbit; cousin)**.** The cyclic group $\mathbb{Z}_n$ acts on size-$(n-3)$ multisets by vertex shift; an *orbit* is a $\mathbb{Z}_n$-equivalence class. Given a Step-3 fingerprint equation $\sum_C \alpha_C\, a_C = 0$ at $(\mathcal{Z}, k, \mathcal{F})$, the multisets $\{C : \alpha_C \neq 0\}$ are the *cousins* of any M that occurs in the equation; their orbit-classes are the *cousin orbits* of $\mathrm{orbit}(M)$.

2 Single-orbit cascade and block-rule cascade

Definition 3 (Single-orbit cascade)**.** Orbit $\mathcal{O}$ admits a *single-orbit depth-1 cascade* recipe iff there is a triple $(\mathcal{Z}, k, \mathcal{F})$ with $k = \ell_{\mathcal{Z}}(M) + 1$ for some $M \in \mathcal{O}$ such that: (i) the fingerprint equation has a_M with nonzero scalar; and (ii) every cousin of M in this equation is Step-1-killable at some cyclic zone.

When (i)+(ii) hold, substituting cousin-zeros into the fingerprint equation collapses it to $\alpha_M\, a_M = 0$, so $a_M = 0$. By cyclic equivariance, the recipe rotates onto every member of $\mathcal{O}$.

The mechanism that closes $n = 7, 8$ is: *every Step-1 survivor orbit admits a single-orbit recipe.* The $n = 9$ data falsifies this for 4 orbits (and 15 timeouts that turn out to lie in the same cluster).

Definition 4 (Coupled cluster)**.** The *coupling graph* G_n has vertex set = Step-1 survivor orbits at n. An edge $\mathcal{O} \sim \mathcal{O}'$ exists iff there is a Step-3 fingerprint equation with $\mathrm{orbit}(M) = \mathcal{O}$ and $\mathrm{orbit}(C) = \mathcal{O}'$ for some cousin C that is itself a Step-1 survivor. A *cluster* is a connected component of G_n.

Definition 5 (Block-rule cascade)**.** Let $\mathcal{C} = \{\mathcal{O}_1, \ldots, \mathcal{O}_m\}$ be a cluster. For each orbit $\mathcal{O}_i$, enumerate all admissible depth-1 fingerprint recipes; collect their fingerprint equations into a matrix $M^{\mathcal{C}}_{\mathrm{FP}}$ with rows indexed by equations and columns by cluster orbits (interior columns) plus external orbits (Step-1 survivors not in $\mathcal{C}$ and triangulation orbits). The cluster is *block-rule killable at depth-1* iff the cluster sub-matrix (interior columns) has full rank $m = |\mathcal{C}|$, with externals at their tree-amplitude values—triangulation columns equated by Step-2 to a common c, non-cluster survivor columns set to 0 by their own single-orbit cascades.

Proposition 6 ($n = 6$ baseline: 4-orbit perfect-matching cluster)**.** *At $n = 6$, after Step-1 and Step-2 exhaust, four perfect-matching orbits survive: $M_A = \{X_{13}, X_{25}, X_{46}\}$, $M_B = \{X_{14}, X_{25}, X_{36}\}$, $M_C = \{X_{14}, X_{26}, X_{35}\}$, $M_D = \{X_{15}, X_{24}, X_{36}\}$. The depth-1 fingerprint matrix $M^{\mathcal{C}}_{\mathrm{FP}}$ over the six cyclic zones is a 6×4 block of rank 3, with nullspace spanned by $(1, -1, 1, 1)$. The residual $r = 1$ is closed by a depth-2 anchor: at $\mathcal{Z}_{1,3}$, the order-X_{13}^1 coefficient of $1/(X_{14}X_{15}^2X_{36})$ uniquely isolates a_{M_B} after Step-1 cousins die at $\mathcal{Z}_{6,3}$. Together, $a_{M_A} = a_{M_B} = a_{M_C} = a_{M_D} = 0$.*

The proof is in `paper/Laurent series for hard kills/cascade_n7.tex` (precursor calculation) and the companion file `fingerprint_lemma_5pt_6pt.tex`. The verification script `verify_6pt_anchor_kill.py` extracts the anchor symbolically.

3 The $n = 9$ result

Step-1 enumeration at $n = 9$ yields 1,011 non-triangulation survivors organized into 113 cyclic orbits (one of size 3, $\mathbb{Z}_3$-stabilised; 112 of size 9, free orbits).

3.1 Single-orbit pass

The script `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` attempts a single-orbit depth-1 recipe for each of the 113 orbit representatives, with a 600-second timeout per orbit:

Outcome		count
single-orbit recipe found		94
search timeout (recipe undetermined)		15
search exhausted, no recipe found	4	$\{\mathcal{O}_{22}, \mathcal{O}_{46}, \mathcal{O}_{88}, \mathcal{O}_{108}\}$

The 4 exhausted-search failures are diagnosed in `.../diagnostics/diagnose_orbit_22.md`; they are not artifacts of bounded depth or pruning. For example, $\mathcal{O}_{22}$ has representative $M_{22} = \{X_{1,3}, X_{1,4}, X_{1,7}, X_{1,8}, X_{4,6}, X_{5,7}\}$ and at every depth $1 \leq k \leq 4$ each fingerprint equation contains at least one cousin that is itself a Step-1 survivor; specifically $\mathcal{O}_{22}$ couples directly to $\mathcal{O}_{88}, \mathcal{O}_{96}, \mathcal{O}_{102}$ at the $\mathcal{Z}_{3,5}$ depth-1 fingerprint $X_{3,6}/(X_{1,3}X_{1,7}X_{1,8}X_{5,7})$.

3.2 Cluster BFS and matrix

A breadth-first traversal of the coupling graph G_9 starting from $\mathcal{O}_{22}$ absorbs 90 orbits (all 4 failures and all 15 timeouts among them). The remaining 23 orbits form a separate connected component (or trivial components), each killed by its single-orbit recipe.

For the 90-orbit cluster $\mathcal{C}_{90}$, the depth-1 cascade matrix tabulates 506 fingerprint equations against 143 columns (90 cluster orbits +32 triangulation orbits +21 non-cluster survivor orbits). Working over $\mathbb{Q}$ via `sympy`:

quantity	value
cluster size $\lvert\mathcal{C}_{90}\rvert$	90
depth-1 fingerprint equations	506
external columns (triangulations + non-cluster survivors)	$32 + 21 = 53$
$\mathrm{rank}(M_{\mathrm{FP}}^{\mathcal{C}_{90}})$ on cluster columns	**90**
cluster residual $r = \lvert\mathcal{C}_{90}\rvert - \mathrm{rank}$	**0**
full-system rank (cluster + externals)	140/143
full-system nullity (located in external columns)	3

Theorem 7 ($n = 9$ block-rule kill)**.** *With externals set to their tree-amplitude values—each of the* 32 *triangulation orbits equal to a common scalar* c *via the Step-2 flip-graph, each of the* 21 *non-cluster survivor orbits equal to* 0 *via its single-orbit cascade—the depth-1 cascade system on* $\mathcal{C}_{90}$ *forces* $a_{\mathcal{O}_i} = 0$ *for every* $\mathcal{O}_i \in \mathcal{C}_{90}$*. In particular, the four single-orbit failures* $\mathcal{O}_{22}, \mathcal{O}_{46}, \mathcal{O}_{88}, \mathcal{O}_{108}$ *are killed jointly with the rest of* $\mathcal{C}_{90}$*.*

The proof is the rank-90 calculation reported above; the trace is `.../n9/outputs/n9_cluster_partial_results.js`
The 3-dimensional full-system nullity reflects the residual freedom in external columns before Step-2 triangulation-equating and single-orbit cascades on the 21 non-cluster orbits are imposed. After all auxiliary equations are added, the nullity collapses to 1 (overall scaling), giving the locality theorem at $n = 9$: $B = c \cdot A_9^{\mathrm{tree}}$.

3.3 Comparison with $n = 6$

	$n = 6$ block	$n = 9$ block
cluster size m	4	90
fingerprint equations	6	506
matrix rank on cluster columns	3	90
residual r	1	0
anchor required?	yes (one at X_*^1)	no

The $n = 6$ cluster had *barely enough* equations to determine itself, leaving a 1-dimensional residual that needed an anchor kill. The $n = 9$ cluster has a $\sim 5\times$ ratio of equations to orbits and is grossly overdetermined. The $n = 6$ "anchor-needed" case is plausibly a low-n artifact: at higher n the fingerprint-equation count grows polynomially with cluster size, and the cluster matrix becomes generically full-rank.

4 Geometric interpretation

The cascade mechanism is the algebraic shadow of a geometric operation. At zone $\mathcal{Z}_{r,r+2}$, the substitution $X_{r+1,k} = X_{r,k} - X_*$ encodes the *ear-collapse* that identifies vertex $r+1$ with the rest of the polygon. The bare relation $X_{r+1,r-1} = -X_*$ is the *core* 4*-gon kinematic identity*: at $n = 4$, the only chord is $X_{1,3} = -X_{2,4}$, and every higher-n collapse uses this identity at one local cyclic position.

The Laurent expansion in $1/X_*$ records the *kinematic memory* of the collapse: each non-bare substitute $X_{r+1,k}$ remembers its companion $X_{r,k}$ at order $X_*^{-(j+1)}$ (coefficient $X_{r,k}^j$). Depth-d thus probes the d-th layer of memory.

A multiset M killable by a depth-d cascade at zone r has d chord-pairs whose crossing structure interacts with the r-vertex collapse. A multiset *not* killable by any single zone's cascade—the cluster case—has crossings that are coupled across multiple cyclic orbits: configurations reachable from each other by a single ear-collapse move. The cluster is the connected component of the *associahedron face graph* restricted to the relevant vertex-merge action.

The fact that the depth-1 system on a cluster has rank $\geq |\mathcal{C}| - 1$ is, geometrically, the statement that a connected face of the associahedron has only finitely many (and typically zero) "free directions" left after all hidden-zero constraints are imposed. The anchor kill at $n = 6$ is the 1-form needed to close the last degree of freedom on a small face; at $n = 9$ the larger face is rigidified by the constraint count alone.

5 Refined Conjecture 10

The original Conjecture 10 of the master plan stated: *every Step-1+2 survivor orbit admits a depth-1 single-orbit cascade recipe.* The $n = 9$ data falsifies this. The corrected conjecture incorporates the cluster mechanism:

Conjecture 8 (Block-rule cascade, refined Item 10)**.** *For every $n \geq 7$ and every cluster $\mathcal{C}$ of Step-1+2 survivor orbits at n:*

(1) *the cluster $\mathcal{C}$ is finite;*

(2) *there exists $r = r(\mathcal{C}) \geq 0$ with* $\operatorname{rank}(M_{\mathrm{FP}}^{\mathcal{C}}) = |\mathcal{C}| - r$*;*

(3) *if $r \geq 1$, the residual r-dimensional space is closed by r higher-Laurent-order anchor equations, each isolating a distinct combination of cluster coefficients with all extra cousins Step-1-killable.*

Corollary 9 (Locality, conditional)**.** *Conjecture 8, together with: the Step-1 exhaustivity of [1, §2]; the triangulation-survives- Step-1 lemma of [2]; the Subset-Vanishing Theorem of [3] (after notes #11, #18); and the Step-2 flip-graph connectivity of the associahedron, implies $B|_{\mathcal{Z}_r} = 0\ \forall r \Rightarrow B = c \cdot A_n^{\mathrm{tree}}$ for all n.*

The conjecture is verified at $n = 6, 7, 8, 9$. At $n = 6$, $r = 1$ with one anchor; at $n = 7, 8$, all clusters have $|\mathcal{C}| = 1$ and $r = 0$; at $n = 9$, the 90-orbit cluster has $r = 0$. The pattern is consistent with the prediction that $r \to 0$ as n grows: the fingerprint-equation count overdetermines large clusters.

6 What remains for the proof

The proof shape is now complete in architecture; remaining work is:

1. **Cluster finiteness**. Every cluster is finite. Likely follows from finite ansatz dimension at fixed n, but should be recorded as a lemma.
2. **Cluster rank lower bound**. Show $\operatorname{rank}(M_{\mathrm{FP}}^{\mathcal{C}}) \geq |\mathcal{C}| - 1$ at all n. This is the core combinatorial-geometric content. A generating-function approach (counting fingerprint equations vs cluster orbits) is plausible.
3. **Anchor existence when** $r = 1$. When the residual is 1-dimensional (the $n = 6$-style case), exhibit a higher-Laurent anchor that closes it. Conjectured to exist generically.
4. **Step-2 flip-graph connectivity**. Show that bare-swap relations across zones realize the full associahedron flip graph. For $n \leq 6$ this is standard; for general n it is likely already in the combinatorics literature on flip graphs.
5. **Subset-vanishing in the non-local cascade**. Notes #11, #18 ([5, 6]) prove the d-subset specialisation of the Subset-Vanishing Theorem rigorously, beyond [7, Appx A]. Generalising the d-subset framework to the cascade setting may yield a structural proof of (2) above.

Files and reproducibility

- `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` — the single-orbit cascade engine.
- `computations/step4_laurent_block_analysis/n9/scripts/` — the cluster-BFS, matrix-rank, and anchor-search scripts.
- `computations/step4_laurent_block_analysis/n9/outputs/` — `n9_cluster_partial_results.json` and `n9_failure_equations_readable.txt`.
- `computations/step4_laurent_block_analysis/n9/diagnostics/` — per-anomaly walkthroughs for orbits $22, 25, 28, 34, 35, 46, 88, 108$.

References

[1] *Step-1 kill statistics and asymptotic locality.* Companion satellite, this repository, `paper/step-one-kill-statistics/`.

[2] *Triangulations are not Step-1 killable.* Companion satellite, this repository, `paper/step-one-doesnt-kill-triangulations/`.

[3] *The Subset-Vanishing Theorem:* $B \equiv 0 \Rightarrow B_i \equiv 0$. Companion satellite, this repository, `paper/B=0->B_i=0/`.

[4] *The Laurent-tail kill at* $n = 7$*: seven layer-*0 *survivors and how they die.* Companion satellite, this repository, `paper/Laurent series for hard kills/`.

[5] Handwritten note #11, *finale-d-subset-induction*, this repository, `notes/`.

[6] Handwritten note #18, *d-subset rigorous proof*, this repository, `notes/`.

[7] L. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory.* arXiv:2406.04234, 2024.